Physics A (H156, H556)

Exploring - Coulombs law - ppq
KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **53**.
- The total number of marks may take into account some 'either/or' question choices.

1 In the Rutherford scattering experiment alpha particles are directed at a gold foil.

Gold nuclei have 79 protons. The distance of closest approach is 47.0 fm.

Which is the best estimate of the work done on an alpha particle as it moves from 53.0 fm to the point of closest approach?

A 10^{-18} J

B 10^{-16} J

C 10^{-15} J

D 10^{-13} J

Your answer [1]

2 The centres of a positron and a helium nucleus are separated by 2 mm.

What is the electrostatic force between them?

A 1.15×10^{-28} N

B 2.30×10^{-25} N

C 5.75×10^{-23} N

D 1.15×10^{-22} N

Your answer [1]

3(a) The diagram below shows two parallel plates, **E** and **C**, in an evacuated glass tube.

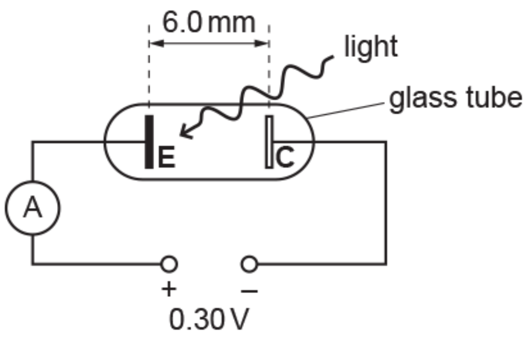


Plate **E** is made from potassium, which is sensitive to light. Plate **C** is not sensitive to light.

The separation between the plates is 6.0 mm and the potential difference between the plates is 0.30 V.

Light of frequency 6.3×10^{14} Hz is incident on plate **E**. The photoelectrons emitted from this plate have **maximum** kinetic energy 0.30 eV (4.8×10^{-20} J). The photoelectrons are repelled by the negative plate **C**. The ammeter reading is zero because these photoelectrons reach plate **C** with zero kinetic energy.

This question is about a photoelectron emitted perpendicular to plate **E** and with an initial kinetic energy of 4.8×10^{-20} J.

i. Show that the magnitude of deceleration of this photoelectron is 8.8×10^{12} ms^{-2}.

[3]

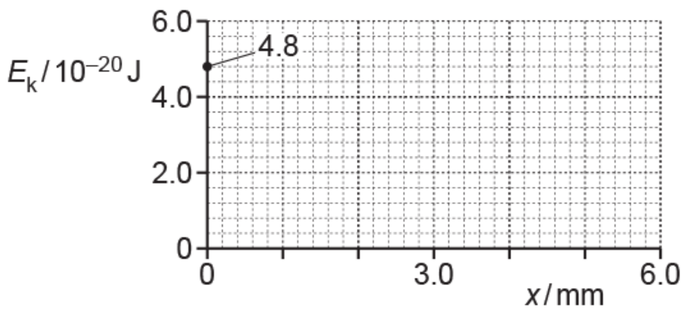
ii. Show that the initial speed of the photoelectron is about 3×10^{5} ms^{-1}.

[2]

iii. Calculate the time *t* taken by the photoelectron to travel from plate **E** to plate **C**.

t = ... s **[2]**

iv. Using the axes shown below, sketch a graph of kinetic energy E_k against distance x from plate **E**.



[2]

(b) Explain, in terms of photons, what happens to the ammeter reading when light of frequency greater than 6.3×10^{14} Hz is now incident on plate **E**.

[2]

4 An electron has both mass and charge. The electron has a gravitational field and an electric field around it.
Which statement is **not** correct?

A Both field patterns look the same.

B Both field patterns show parallel field lines around the electron.

C Both field strengths obey an inverse square law with distance from the electron.

D The direction of both fields is the same at any point around the electron.

Your answer **[1]**

5 The diagram below shows two oppositely charged spheres.

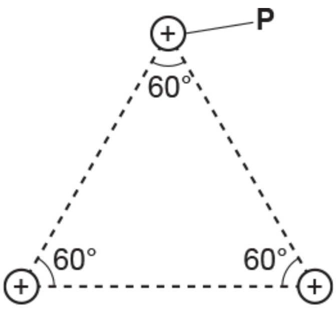
The magnitude of the charge on each sphere is the same.
The point **P** is on the line joining the centres of the spheres and is the same distance from the centre of each sphere.

Which statement is correct?

A A negatively charged particle at **P** will move to the right.
B The direction of the electric field at **P** is to the left.
C The electric potential at **P** is zero.
D The magnitude of the electric field strength at **P** is zero.

Your answer ☐ **[1]**

6 The diagram below shows the arrangement of the 3 protons inside the nucleus of lithium-6 ($^{6}_{3}Li$).



The separation between each proton is about 1.0×10^{-15} m.

i. Calculate the magnitude of the repulsive electric force *F* experienced by the proton **P**.

F = ... N **[4]**

ii. On the diagram above, draw an arrow to show the direction of the electric force *F* experienced by **P**.

[1]

iii. Explain how protons stay within the nucleus of lithium-6.

....

....

....

.... **[2]**

7 The electric field strength at a distance of 2.0×10^{-8} m from a nucleus is 3.3×10^{8} N C^{-1}.

What is the charge on the nucleus?

A 1.6×10^{-19} C

B 1.5×10^{-17} C

C 7.3×10^{-10} C

D 3.8×10^{-9} C

Your answer **[1]**

8 Which law indicates that charge is conserved?

A Lenz's law

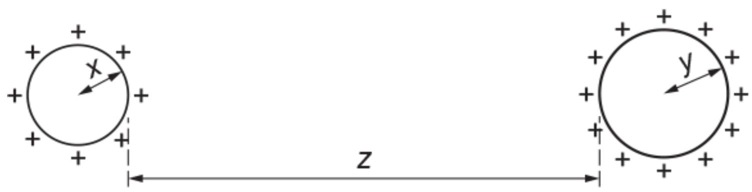
B Coulomb's law

C Kirchhoff's first law

D Faraday's law of electromagnetic induction

Your answer **[1]**

9 The diagram below shows two uniformly charged spheres separated by a large distance *z*.



The radius of the small sphere is x and the radius of the large sphere is *y*.

Which is the correct distance to use when determining the electric force between the charged spheres?

A z

B $x + z$

C $y + z$

D $x + y + z$

Your answer [1]

10

The nuclear reaction below shows how the isotope of fluorine-18 ($^{18}_{9}F$) is made from the isotope of oxygen-18 ($^{18}_{8}O$) .

$$^{18}_{8}O + ^{1}_{1}p \rightarrow ^{18}_{9}F + ^{1}_{0}n + \gamma$$

The oxygen-18 nucleus is **stationary** and the proton has kinetic energy of 0.25×10^{-11} J.

The binding energy of the $^{18}_{8}O$ nucleus is 2.24×10^{-11} J and the binding energy of the $^{18}_{9}F$ nucleus is 2.20×10^{-11} J. The proton and the neutron have zero binding energy.

i. Explain why a high-speed proton is necessary to trigger the nuclear reaction shown above.

-- [2]

ii. Estimate the minimum wavelength λ of the gamma ray photon (γ).

λ = -------------------------------- m [3]

iii. Fluorine-18 is a positron emitter.

Name a medical imaging technique that uses fluorine-18 and state one benefit of the technique.

-- [2]

11 Fig. 21.1 shows two oppositely charged ions to the left of a point **X**.

Fig. 21.1

The separation between the centres of the ions is 3.0×10^{-10} m. Each ion has charge of magnitude 1.6×10^{-19} C.

i. Explain why the direction of the **resultant** electric field strength at point **X** is to the left.

____

____

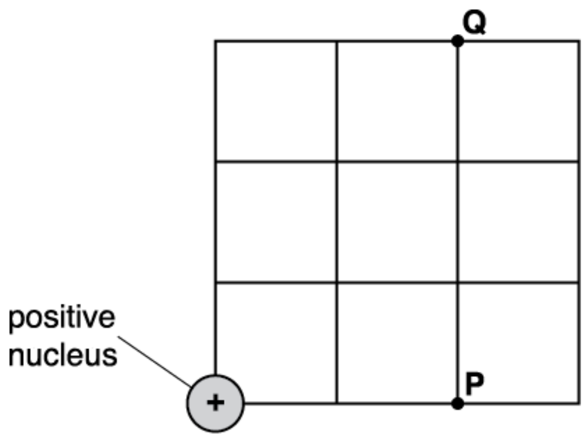
____

____ **[2]**

ii. Calculate the minimum energy in eV required to completely separate the ions.

energy = ______________________________ eV **[3]**

12 An electron at point **P** experiences an electric force of magnitude 1.8 μN due to the positive nucleus.



What is the magnitude of the force experienced by the same electron when it is at point **Q**?

A 0.28 μN
B 0.55 μN
C 1.0 μN
D 1.8 μN

Your answer

[1]

13 Fig. 21.2 shows a uniformly charged sphere of radius R.

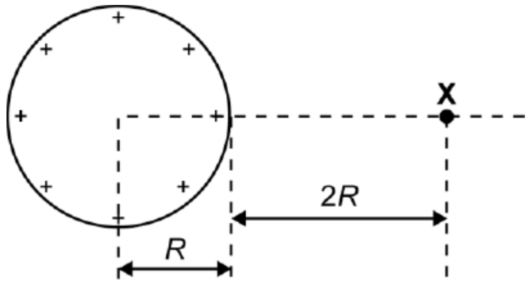


Fig. 21.2

The electric potential at point **X** is +1800 V. Point **X** is at a distance of $2R$ from the **surface** of the sphere.

i. Calculate the electric potential V at the surface of the sphere.

V = ____________________ V **[2]**

ii. The radius of the sphere is 4.0 cm.

Calculate

1. the surface charge Q on the sphere

Q = ____________________ C **[2]**

2. the electric field strength E at the surface of the sphere.

E = ____________________ N C^{-1} **[2]**

14(a) Describe the similarities and the differences between the gravitational field of a point mass and the electric field of a point charge.

--- [3]

(b) **Fig. 21.1** shows two identical negatively charged conducting spheres.

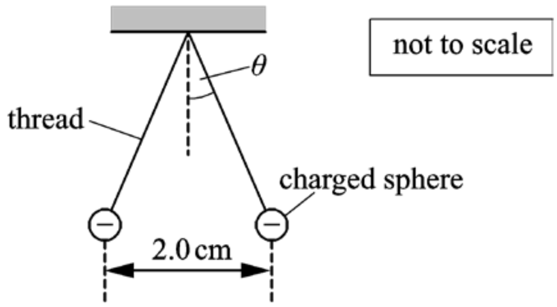


Fig. 21.1

The spheres are tiny and each is suspended from a nylon thread. Each sphere has mass 6.0×10^{-5} kg and charge -4.0×10^{-9} C. The separation between the centres of the spheres is 2.0 cm.

i. Explain why the spheres are separated as shown in **Fig. 21.1**.

--- [2]

ii. Calculate the angle θ made by each thread with the vertical.

θ = ________________ ° **[4]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1			D	1	<u>Examiner's Comments</u> A little under half of the candidates were able to calculate and select the correct response. The most common incorrect response was **B**, although it was not entirely clear why this was the case from candidates working. A very small minority of candidates did not answer this question.
			Total	1	
2			D	1	<u>Examiner's Comments</u> Around two thirds of candidates were able to correctly apply Coulomb's law and obtain the correct response. Response **C** was a common distractor and this was evident for working as the charge on the helium nucleus was given as a single multiple of the fundamental charge.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
3	a	i	$a = \frac{VQ}{dm}$ OR $a = \frac{EQ}{m}$ OR $KE = ½ mv^2$ and $v^2 = u^2 + 2as$ OR $KE = F \times d$ and $F = m \times a$ $a = \frac{0.30 \times 1.6 \times 10^{-19}}{6.0 \times 10^{-3} \times 9.11 \times 10^{-31}}$ / $a = \frac{50 \times 1.6 \times 10^{-19}}{9.11 \times 10^{-31}}$ / (Use of $KE = ½ mv^2$) = $4.8 \times 10^{-20} = ½ \times 9.11 \times 10^{-31} \times v^2$ and (use of $v^2 = u^2 + 2as$ =) $v^2 = (1.05 \times 10^{11}) = 2 \times a \times 6 \times 10^{-3}$ ($\pm 0^2$) (Use of $KE = F \times d$) = $4.8 \times 10^{-20} = F \times 6 \times 10^{-3}$ and (use of $F = m \times a$) $F = (8.0 \times 10^{-18}) = 9.11 \times 10^{-31} \times a$ $a = 8.78 …. \times 10^{12}$ (ms^{-2})	C1 M1 A1	**Allow** u and v interchangeably throughout **Allow** calculation of E = (0.30 / 6 × 10^{-3}) = 50 (V m^{-1}) or $v = 3.2 \times 10^5$ (ms^{-1}) or $v^2 = 1.05 \times 10^{11}$ $(ms^{-1})^2$ or $F = 8.0 \times 10^{-18}$ (N) for C1 mark Substitution mark – in any arrangement. Expect full substitutions including numerical value of m_e if appropriate Method 1: direct calculation of a Method 2: using $KE = ½ mv^2$ and $v^2 = u^2 + 2as$ Method 3: using $KE = F \times d$ and $F = m \times a$ **Note** must be more than 2 SF (not paper SF penalty) **Ignore** negative sign <u>Examiner's Comments</u> There were many different routes to showing the acceleration, and marks were given for each method or part method. No one method was seen significantly more than others, and some candidates used a variety of pathways to come to their answer. The main principle in the question (and the subsequent one) where the candidate is being asked to "show that" a given value is correct is that the examiner must be convinced that the candidate has clearly demonstrated that they have carried out the calculation and evaluated it on their calculator. The instructions which examiners used was: first marking point for providing one (or two) equations that would lead to the solution, or calculation of an intermediate value; second marking point for a full substitution into one or more equations; third marking point for using this full substitution to produce an answer to more sf than given in the question. As the second marking point was deemed to be an M mark, the full substitution needed to be seen to gain the A mark. A small number of (often higher end)

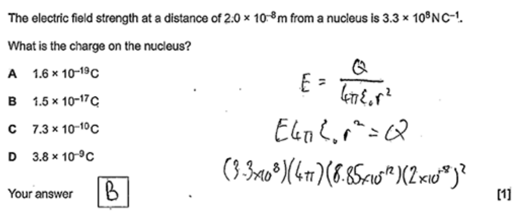
Question			Answer/Indicative content	Marks	Guidance
					candidates did not show the full substitution, often missing out the value of m_e in their calculation, and another common error was to not show the extra significant figure. Over half of the candidates were able to achieve full marks on this question and it generally discriminated well. **Assessment for learning** When a question asks a candidate to "show that" a given value is correct, the following two points should be considered: • Each stage of the calculation should be clearly shown. Preferably setting out any equation first, and then showing a full substitution of all values into that equation • If the value calculated by the candidate would correctly round to the given value, then the candidate should show their calculated value to at least one more significant figure than the given value. Both of these are evidence that the complete calculation has taken place and that the candidate has not somehow artificially generated the required value. This advice should be viewed as "best practice" rather than a rigid set of rules. Reverse arguments are often possible where a candidate can work backwards from their given value, however this is not the advised approach.

Question			Answer/Indicative content	Marks	Guidance
		ii	(Use of $KE = ½\ mv^2$) = $4.8 \times 10^{-20} = ½ \times m \times v^2$ OR ($u^2 = v^2 - 2as$) = $0^2 - [2 \times (-)\ 8.8 \times 10^{12} \times s]$ Full substitution leading to $v = 3.2… \times 10^5$ (ms^{-1})	C1 A1	**Allow** u and v interchangeably Numerical value of m_e must be used if using *KE* method **Note** must be more than 1 SF (not paper SF penalty) **Note** 3.25 is acceptable for A1, but not 3.3 <u>**Examiner's Comments**</u> The vast majority of candidates were able to clearly show that the speed of the photoelectron could be calculated as $3.2 \times 10^5\ ms^{-1}$, most often through substitution into the kinetic energy formula. As in Question 19 (b) (i), it is important to show all variables and constants used in the equation for full marks and to give the answer to at least one more d.p. than given in the question, to show the calculation has taken place. An alternative solution using an equation of motion and the acceleration given (or calculated) in Question 19 (b) (i) would yield the same result.
		iii	$t = \frac{3.2\ \times 10^5}{8.8 \times 10^{12}}$ $t = 3.6 \times 10^{-8}$ (s)	C1 A1	**Allow** correct full substitution into any suvat equation **Allow** 3×10^5 for v **Ignore** signs of substituted values **Expect** values between 3.4×10^{-8} and 3.8×10^{-8} (s) **No ecf** from (b)(i) or (b)(ii) <u>**Examiner's Comments**</u> Only around half of the candidates were able to obtain answers within the required range. Candidates used a variety of rounded or none-rounded values from prior calculations, so a generous range of responses was given to allow for this. A common error among less successful responses was to simply use speed = distance/time usually leading to 2.0×10^{-8} s. Those using $s = ut + ½\ at^2$ often encountered problems in solving the equation as it could lead to imaginary roots and tended to produce solutions way outside of the accepted values. However, marks could be given for setting up the calculation correctly.

Question			Answer/Indicative content	Marks	Guidance
		iv	Any line with negative slope starting from 0, 4.8×10^{-20} A straight line finishing at 6.0,0	M1 A1	½ square tolerance. **ALLOW** a curve with negative gradient. **ALLOW** a region of zero gradient, but not whole line ½ square tolerance on axis intercepted Examiner's Comments Nearly all candidates appreciated that this line would start at 4.8×10^{-20} J and decrease to zero at 6.0 mm. However, the vast majority drew a curved line of decreasing gradient. This may well have come from a confusion from $KE \propto v^2$ and attempting to draw a parabola.
	b		Energy of photon increases (max) kinetic energy / speed (of electrons) increases / (some) electrons (now) reach C **and** there is a current or reading (on ammeter)	B1 B1	**Do not allow** increased *kinetic* energy of photons **Do not allow** explanation in terms of increased number of **emitted** electrons (per second) **Allow** photoelectrons for electrons Examiner's Comments There were several misconceptions in candidates' responses to this question. Many candidates did not appreciate that the increased frequency would result in electrons of greater KE and thought that it was the increased energy of the photons crossing the 6.0mm gap that caused an ammeter reading. A significant number of candidates also described increasing frequency causing an increase in *kinetic* energy of photons, and some also linked the increasing frequency to a greater number of photons or photoelectrons.
			Total	11	
4			B	1	Examiner's Comments This was a challenging question; candidates often drew a radial field diagram as a starting point and many candidates helpfully "ticked" the correct responses to help them eliminate these from their response. It is important that candidates read the questions carefully, as it was evident that some were looking for a correct statement.

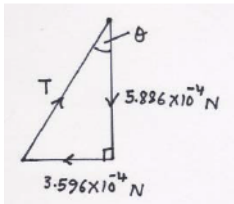
Question			Answer/Indicative content	Marks	Guidance
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
5			C	1	<u>Examiner's Comments</u> The correct response is **C**. Candidates often find the electric field questions challenging and this was again the case as this question was correctly answered by only one third of the candidates. Many candidates drew arrows on the diagram to assist them. Response **D** was the most common distractor; linking this to gravitational fields would produce a zero field strength at **P** which is likely to be the reason.
			Total	1	
6		i	(force =) $\frac{(1.6\times10^{-19})^2}{4\pi\epsilon_0\times(1.0\times10^{-15})^2}$ (F =) 230 (N) $F^2 = 230^2 + 230^2 - 2\times230\times230\times\cos120°$ **or** $F = 2\times230\cos30°$ F = 400 (N)	C1 C1 C1	**Special case:** $F = \frac{Qq}{4\pi\epsilon_0 r^2} = \frac{2\times1.6\times10^{-19}}{4\pi\epsilon_0\times(1.0\times10^{-15})^2}$ loses this C1 mark, then ECF for the rest of the marks **Not** the first two C1 marks for incorrect charge, then allow ECF for the final C1A1 marks **Note** force to 4 SF is 230.2 N **Allow** sine rule / scale drawing **Allow** this mark for 230cos30° **or** 200 (N) **Allow** ± 10 (N) if scale drawing used
		ii	F / arrow vertical up the page	B1	**Allow** correct arrow direction anywhere on the figure
		iii	Strong (nuclear) force (acts on the protons) The strong (nuclear) force is attractive	B1 B1	**Ignore** gravitational force **Allow** pulls / holds (the protons) / binds (the protons) for 'attractive'
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
7			B	1	**Examiner's Comments** This question required knowledge of the equation $E = \frac{Q}{4\pi\varepsilon_0 r^2}$, which is in the Data, Formulae and Relationship Booklet. The key, the correct answer, is **B**. The most common mistake made by candidates was not squaring *r*, or the equivalent where the electric potential *V* equation was used instead. This gave the incorrect answer of 7.3 × 10^{-10} C for the most recurrent distractor **C**. The exemplar 2 below shows a plausible method for getting to the correct answer. **Exemplar 2**  The electric field strength at a distance of 2.0 × 10^{-8} m from a nucleus is 3.3 × 10^{8} N C^{-1}. What is the charge on the nucleus? **A** 1.6 × 10^{-19} C **B** 1.5 × 10^{-17} C **C** 7.3 × 10^{-10} C **D** 3.8 × 10^{-9} C Your answer B [1] $E = \frac{Q}{4\pi\varepsilon_0 r^2}$ $E 4\pi \varepsilon_0 r^2 = Q$ $(3.3\times10^{8})(4\pi)(8.85\times10^{-12})(2\times10^{-8})^2$ This middle-grade candidate has shown all the working. A significant number of candidates wrote down the correct expression, circled the key data in the question and did the rest of the work on their calculators. A variety of techniques were use.
			Total	1	
8			c	1	**Examiner's Comments** This was a well-answered question with most candidates correctly recalling that charge is conserved according to Kirchhoff's first law. A significant number of candidates distracted towards **B**; perhaps because of the unit of charge is the coulomb.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
9			D	1	<u>Examiner's Comments</u> This question was assessing the simple learning outcome 6.2.1b from the specification – *modelling a uniformly charged sphere as a point charge at its centre*. The correct key is **D**. The most popular distractor was **A** with distance being measured between the two surfaces of the sphere.
			Total	1	
10		i	Proton is repelled (by nucleus)	B1	
			(High-speed) proton can get close to (oxygen) nucleus	B1	**Allow** 'proton can experience the strong (nuclear) force' **Not** 'collide / hit nucleus'
		ii	E = [0.25 - (2.24 - 2.20)] × 10^{-11} (J) **or** 0.21 × 10^{-11} (J)	C1	
			$\lambda = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{0.21 \times 10^{-11}}$ (Any subject)	C1	
			λ = 9.5 × 10^{-14} (m)	A1	**Allow** 2 marks for 6.9 × 10^{-14}; E = 0.29 × 10^{-11} used **Allow** 1 mark for a value correctly calculated based on any other incorrect value for E (e.g. 8(.0) × 10^{-14} for E = 0.25 × 10^{-11} and 5(.0) × 10^{-13} for E = 0.04 × 10^{-11})
		iii	Used in PET (scans)	M1	Enter text here.
			Any <u>one</u> from: Used to diagnose function of organ / brain / body Detection of cancer / tumour Non-invasive / no surgery / no infection 3D (image)	A1	
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
11		i	The direction of the electric field due to the negative charge is to the left and right for the positive charge.	B1	
		i	The magnitude of the electric field strength due to the positive charge is smaller than that for the negative charge (because of greater distance). (Hence the resultant electric field strength is to the left.)	B1	
		ii	energy = $\dfrac{Qq}{4\pi\varepsilon_0 r} = \dfrac{(1.60\times10^{-19})^2}{4\pi\varepsilon_0 \times 3.0\times10^{-10}}$	C1	
		ii	energy = 7.67(2) × 10^{-19} (J)	C1	
		ii	energy = 4.8 (eV)	A1	
			Total	**5**	
12			B	1	
			Total	**1**	
13		i	$V \propto 1/r$ or distance = 3R	C1	
		i	V = 5400 (V)	A1	
		ii	**1** $5400 = \dfrac{Q}{4\pi\times8.85\times10^{-12}\times0.04}$ (Any subject)	C1	Possible ecf from **(i)**
		ii	Q = 2.4 × 10^{-8} (C)	A1	
		ii	**2** $E = \dfrac{2.4\times10^{-8}}{4\pi\times8.85\times10^{-12}\times0.04^2}$	C1	Possible ecf from **(ii)1**
		ii	E = 1.35 × 10^5 (N C^{-1})	A1	
			Total	**6**	

Question			Answer/Indicative content	Marks	Guidance
14	a		**Similarity** The field strength or force ∝ 1 / separation^2 or both produce a radial field.	B1	
			Differences Gravitational field is linked to mass and electric field is linked to charge.	B1	
			Gravitational field is always attractive whereas electric field can be either attractive or repulsive.	B1	
	b	i	The charges repel each other (because they have like charges).	B1	
		i	Each charge is in equilibrium under the action of the three forces: downward weight, a horizontal electrical force and an upwardly inclined force due to the tension in the string.	B1	
		ii	$F = \dfrac{(4.0\times10^{-9})^2}{4\pi\varepsilon_0\times0.02^2} = 3.596....\times10^{-4}$ (N)	C1	Correct use of $F = \dfrac{Qq}{4\pi\varepsilon_0 r^2}$
		ii	weight $W = 6.0 \times 10^{-5} \times 9.81 = 5.886 \times 10^{-4}$ (N)	C1	
		ii	$\tan\theta = \dfrac{3.596\times10^{-4}}{5.886\times10^{-4}}$	C1	
		ii	angle $\theta = 31°$	A1	
			Total	**9**	